YASHRAJSINH CHUDASAMA

Senior PhD Researcher | Research Associate

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Academic Profile

Senior PhD researcher and Research Associate at TIB – Leibniz Information Centre for Science and Technology in Hannover, Germany. Investigating explainable causal models over cognitive knowledge graphs, with a focus on integrating semantic representations, symbolic reasoning and machine learning to improve the interpretability and reliability of AI systems.

Research Experience

Research Associate (TrustKG Project) Nov 2021–Present
TIB – Leibniz Information Centre for Science and Technology | Hannover, Germany

- Research explainable knowledge-graph completion, interpretable link prediction and neuro-symbolic reasoning.
- Develop methods connecting symbolic paths, rules, ontologies and language models for evidence-grounded explanations.
- Contribute to scientific knowledge management, semantic machine-learning representations and peer-reviewed publications.

Student Research Assistant Dec 2017–Apr 2018
Sensor data and Semantic Web project

- Developed sensor-data collection, mappings and SPARQL-based querying components using C# and Arduino IDE.

Research Projects and Contributions

Knowledge Graph Explanation and Neuro-Symbolic AI
2021–Present | Symbolic link-prediction explanations, knowledge graph embeddings, rule-guided reasoning and causal models. Projects: InterpretME, NeSyCoT, PRISM, MetaMine and Tool4Boxology.

Nonlinear Semitrailer Dynamics Using RNNs
Sep 2020–Mar 2021 | *M.Sc. thesis, Leibniz Universität Hannover*. Simulated vehicle dynamics in Simulink; generated datasets and trained/evaluated recurrent neural networks with MATLAB, Python and TensorFlow.

Machine Fault Detection with Machine Learning
Feb–Aug 2019 | *M.Sc. thesis, Peter the Great St. Petersburg Polytechnic University*. Sensor-signal segmentation, feature extraction and tool-wear classification across drilling, milling and turning processes.

Electrooculography Using CNNs
Jan 2018 | Convolutional neural networks for bio-limb electrooculography applications (MATLAB/Python).

Education

PhD – in progress
Leibniz Universität Hannover, Germany
Dissertation: "Explainable Causal Models on top of Cognitive Knowledge Graphs."
Start date: April 2022

M.Sc., International Mechatronics
Leibniz Universität Hannover, Germany
2017–2021 | Grade: 1.9

M.Sc., Intelligent Systems
Peter the Great St. Petersburg Polytechnic University, Russia
2017–2019 | Grade: 1.3

B.E., Mechatronics Engineering
Gujarat Technological University, India
2013–2017 | Grade: 1.8

Research Areas

Knowledge Graphs
Explainable AI
Neuro-Symbolic AI
Causal Reasoning
Knowledge Graph Completion
Recurrent Neural Networks
Sensor-based Fault Diagnosis
Semantic Web
Symbolic Reasoning

Technical Expertise

Programming: Python, SPARQL, SQL, MATLAB, C/C++, C#
ML: TensorFlow, Keras, scikit-learn, pandas, torch, transformers, neural nets
Methods: KG embeddings, rules, RNNs, signal processing, RAG, fine-tuning

Publications

2026 **Y. Chudasama**, D. Purohit, E. Iglesias, A. Sakor and M.-E. Vidal. “Generating Symbolic Chain-of-Thoughts from Knowledge Graphs to Fine-Tune Language Models for Dataset Metadata Extraction in Scientific Publications.” Under Review *ARRAY Journal*.

2026 **Y. Chudasama**, D. Purohit and M.-E. Vidal. “PRISM: Path-guided Reasoning and Interpretation for Semantic Models.” Under Review *Expert Systems with Applications Journal*.

2026 **Y. Chudasama**, D. Purohit, M. Forghani, E. Iglesias, ME. Vidal. From Boxes to Knowledge: Tool4Boxology for Semantic Modeling and Analysis of Neuro-Symbolic AI Systems. ISWC 2026.

2026 **Y. Chudasama**, D. Purohit, E. Iglesias, M. Forghani, ME. Vidal. Neuro-Symbolic Design Patterns: A Semantic Representation of Neuro-Symbolic Architectures. ISWC 2026.

2026 J.E. Bendler, **Y. Chudasama**, M. Forghani, E. Iglesias, D. Purohit, J. R. Roney, A. T. Teije, F. van Harmelen, ME. Vidal. Demonstrating Tool4Boxology: Constructing and Analysing Neuro-Symbolic Architectures, In The 38th Benelux Conference on Artificial Intelligence and the 35th Belgian Dutch Conference on Machine Learning, 2026

2026 ME. Vidal, **Y. Chudasama**, E. Iglesias, D. Purohit. From Normal Forms to Knowledge Graph Ecosystems: Foundations for Trustworthy AI. In: Vargas-Solar, G., Stefanidis, K., Palpanas, T., Marcel, P., Halfeld-Ferrari, M. (eds) Advances in Databases and Information Systems. ADBIS 2026. Lecture Notes in Computer Science, vol 16937. Springer, Cham.

2026 J.E. Bendler, **Y. Chudasama**, M. Forghani, E. Iglesias, D. Purohit, J. R. Roney, A. T. Teije, F. van Harmelen, ME. Vidal. “Tool4Boxology: A Semantic Toolbox for Constructing and Analysing Neuro-Symbolic Architectures.” *European Semantic Web Conference (ESWC)*, pp. 191–211.

2025 **Y. Chudasama**, D. Purohit, P. D. Rohde, J. Gercke and ME. Vidal. “InterpretME: A Tool for Interpretations of Machine Learning Models over Knowledge Graphs.” *Semantic Web*, 16(2).

2025 ME. Vidal, **Y. Chudasama**, H. Huang, D. Purohit and M. Torrente. “Integrating Knowledge Graphs with Symbolic AI: The Path to Interpretable Hybrid AI Systems in Medicine.” *Journal of Web Semantics*, 84, 100856.

2025 **Y. Chudasama**, H. Huang, D. Purohit and ME. Vidal. “Toward Interpretable Hybrid AI: Integrating Knowledge Graphs and Symbolic Reasoning in Medicine.” *IEEE Access*, 13, 39489–39509.

2025 D. Purohit, **Y. Chudasama** and ME. Vidal. “Enhancing Medical Knowledge Discovery: A Neuro-Symbolic System for Inductive Learning over Medical KGs.” *WSDM*, pp. 1108–1109.

2025 D. Purohit, **Y. Chudasama** and ME. Vidal. “VANILLA: Validated Knowledge Graph Completion—A Normalization-Based Framework for Integrity, Link Prediction, and Logical Accuracy.” *Knowledge-Based Systems*, 325, 113939.

2025 K. Amouzouvi, **Y. Chudasama**, D. Purohit, et al. “SimE: A Knowledge Graph Embedding Model to Encode Self-Similar Structures Through Algebraic and Geometric Transformations.” *IEEE Access*.

2025 D. Purohit, **Y. Chudasama** and ME. Vidal. “Capturing Symbolic Knowledge of Constraints and Incompleteness to Guide Inductive Learning in Neuro-Symbolic Knowledge Graph Completion.” *K-CAP*, pp. 111–118.

2025 **Y. Chudasama**, D. Purohit, P. D. Rohde, J. Gercke, ME. Vidal. InterpretME: A tool for interpretations of machine learning models over knowledge graphs. In Semantic Web Journal, volume 16, issue 2, pages SW-233511.

2025 R. Abu Ahmad et al. (including **Y. Chudasama**). “Semantic Web and Creative AI—A Technical Report from ISWS 2023.” *arXiv preprint* 2501.18542.

2024 **Y. Chudasama**, D. Purohit, P. D. Rohde, E. Iglesias, M. Torrente and ME. Vidal. “Semantically Describing Predictive Models for Interpretable Insights into Lung Cancer Relapse.” *SEMANTiCS*.

2024 M. Russo, **Y. Chudasama**, D. Purohit, S. Sawischa and ME. Vidal. “Employing Hybrid AI Systems to Trace and Document Bias in ML Pipelines.” *IEEE Access*, 12, 96821–96847.

2024 D. Purohit, **Y. Chudasama**, M. Torrente and ME. Vidal. “VISE: Validated and Invalidated Symbolic Explanations for Knowledge Graph Integrity.” *EXPLIMED @ ECAI*.

2023 **Y. Chudasama**. “Exploiting Semantics for Explaining Link Prediction over Knowledge Graphs.” *ESWC*, pp. 321–330.

2023 **Y. Chudasama**, D. Purohit, P. D. Rohde and ME. Vidal. “Enhancing Interpretability of Machine Learning Models over Knowledge Graphs.” *SEMANTiCS Posters & Demos*.

2023 D. Purohit, **Y. Chudasama**, A. Rivas and M.-E. Vidal. “SPaRKLE: Symbolic caPtuRing of Knowledge for Knowledge Graph Enrichment with LEarning.” *K-CAP*, pp. 44–52.

Academic Service, Mentorship and Reviewing

Mentor – Doctoral Symposium 2026
European Semantic Web Conference (ESWC 2026)
Supported doctoral researchers through academic mentorship and research discussions at the ESWC 2026 Doctoral Symposium.

Mentor – Knowledge Graphs for Water Management Summer School 2026
Summer school on knowledge graphs, semantic modelling and water management
Mentored participants exploring semantic representations and knowledge-graph approaches for water-related research problems.

Conference Reviewing

Peer reviewer for the following international conferences:

- **International Semantic Web Conference (ISWC):** 2025, 2026.
- **European Semantic Web Conference (ESWC):** 2025, 2026.
- **SEMANTiCS Conference:** 2025, 2026.

Additional Journal Reviewing

Knowledge-Based Systems; *Journal of Web Semantics*; Springer Nature Snapp; *Informatics in Medicine Unlocked*; and *Future Generation Computer Systems*.

Academic Supervision

Master's Theses (6) and Bachelor's Theses (3)

No.	Student name	Status	Thesis title
1	Liedtke, Meike	Submitted	A Hybrid AI Method to Explain the Behavior of Link Prediction Models
2	Zhao, Huaxia	Submitted	Documenting Knowledge Graph Embedding and Link Predictions using Knowledge Graphs
3	Arnous, Abdulrehman	Submitted	Enhancing Interpretability of Machine Learning Models for Survival Analysis
4	Yousefian, Behnoosh	Submitted	Exploring the Synergy of Numerical and Symbolic Learning with Shape-Based Constraints
5	Forghani, Mahsa	In progress	Semantic-Driven Techniques for Enhancing the Quality of LLM-Generated Educational Videos for Cardiovascular Diseases.
6	Wolter, Steffan	In Progress	Semantic-Driven RAG Techniques for Enhancing the Quality of LLM-Generated Response for Cardiovascular Diseases.
7	Luttmann, Max	Submitted	Bridging Ontologies and Symbolic Learning: Toward Constraint-Aware Pattern Extraction
8	Torke, Fynn	In progress	Evaluating Different Knowledge Graph Access Strategies for Constraint- and Ontology-Aware Rule Mining
9	Steluschat, Simon	In progress	FedTool4Boxology: A Federated Framework for Specifying Neuro-Symbolic AI Extracted from Scientific Literature

International Academic Training

International Semantic Web Summer School (ISWS) May 2023
Bertinoro, Italy
Attended the international summer school focused on Semantic Web research, linked data, and knowledge representation.

Research Profiles

Google Scholar | DBLP | Personal website.